Utsav Bhardwaj

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Education

Bharati Vidyapeeth's College of Engineering **New Delhi, India**
*B.Tech in Computer Science and Engineering – **SGPA: 9.11*** *Nov 2022 – June 2026*

Work Experience

DRDO (Defence Research and Development Organisation) **Intern – Ongoing**

SSPL – New Delhi

- Engineered a centralized project management system using Next.js, Node.js, and PostgreSQL with **JWT-based RBAC**, supporting 70+ scientists across 9 projects and managing **35+ documents with real-time**
- Built an internal HR portal to digitize paper-based workflows, **managing 200+ intern** applications with centralized tracking and offline/LAN-based access without internet dependency

PROJECTS

HiRest – Real-Time Image Sharpening Model | PyTorch, CUDA, Adam Optimizer **[Jun 2025]**

* Co-developed **HiRest**, a real-time image sharpening framework using knowledge distillation, designing and training a lightweight CNN (**HINet**) distilled from Restormer that achieved **32.81 dB PSNR**, **0.904 SSIM**, and **30+ FPS** at 720p while reducing the model size to ∼**3.5M parameters** (**10× smaller** than the teacher model).
* Implemented dual-supervision training combining pixel loss and distillation loss with HIN blocks, residual decoders, CSFF, and SAM modules for computational efficiency and high-quality image restoration.
* Validated model performance using benchmark datasets and MOS evaluation (**4.1/5 vs 4.3/5** teacher), demonstrating transformer-level perceptual quality on consumer GPUs.

Pariksha AI – RAG based Exam Preparation Platform | ChromaDB, API, OCR **[March 2026]**

* Developed an **AI-powered exam preparation pipeline** using **ChromaDB**, **Claude API**, and **Tesseract OCR**, where NCERT PDFs are chunked, embedded, and retrieved via **dual RAG queries** to generate CBSE-style questions and evaluate handwritten answers, reducing generation time to under 5 seconds per batch.
* Designed a **vector search system** using **ChromaDB** and **all-MiniLM-L6-v2** embeddings with metadata filters across question type, Bloom's taxonomy level, and chapter, enabling style-calibrated question generation with over 0.80 similarity to real board exam patterns.
* Implemented a **two-stage answer evaluation workflow** combining **Tesseract OCR** for handwritten text extraction and **Claude API** for scoring against RAG-retrieved NCERT key points, delivering structured JSON feedback across 6 rubric dimensions within 8 seconds per submission.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL
AI/ML & GenAI: LLMs, Prompt Engineering, RAG, NLP, Transformers, Embeddings, Model Evaluation
Libraries & Tools: NumPy, Pandas, Scikit-learn, PyTorch, Hugging Face, LangChain
Data Techniques: Data Preprocessing, Feature Engineering, Machine Learning, Deep Learning
DevOps & Cloud: Git, REST APIs, AWS (familiar)
Core Subjects: DBMS, Operating Systems, OOPs, Computer Networks

CERTIFICATIONS

- Generative AI Fundamental By Databricks
- SQL Intermediate By HackerRank

ACHIEVEMENTS

- Secured **3rd place** in Web Development Hackathon organized by GDSC BVCOE
- Solved **300+ Questions** of Data structures and algorithm across different platform